



CIT-PEELS NATURALS

**"Turning Waste into Wealth,
Naturally."**

How we came across this problem ?

I read a WHO report stating

indoor air pollution is 3-4 times worse than outdoor air,

which shocked me since we spend most of our time indoors. Curious about

how people cleaned in the past, I asked my grandmother, and she

mentioned using a plant called **optiva grevia**.

Inspired by this, I experimented with waste citrus peels and developed

CIT-PEELS—a natural, non-toxic cleaner as effective as chemical ones

aiming to make homes healthier and the planet cleaner.

USER PERSONA:

MODERN URBAN
MOM

NAME: NEHA KAPOOR

AGE: 30

LOCATION: URBAN INDIA

OCCUPATION: MARKETING
PROFESSIONAL

MARITAL:
STATUS Married

KIDS: 1

GOALS

House Help's Resistance:

Her house help refuses to use natural cleaners, claiming they are not as effective and take longer to clean.

Health Concerns:

She worries about exposing her family to toxic chemicals, especially her toddler, who crawls on the floors.

PAIN
POINTS

"I want my house to be clean and safe for my child, but it's so hard to find a natural cleaner that works well and convinces my house help to use it."

QUOTE

Keep her house spotless and safe for her family.
Find a natural cleaner that convinces her house help with its performance.

PROBLEM

1

Toxic Chemicals, VOC's present in market cleaners Available cleaners in the market indeed clean well but they are filled with toxic chemicals, which from direct exposure as well as indirect exposure through residues left on surfaces can cause respiratory issues.

2

Approximately one-third of every citrus fruit is discarded as waste. India generates 10 million tons of citrus waste each year.





PoC & TRL

PoC Achievements:

- Developed stable, antimicrobial, 100% natural cleaners from citrus waste.
- Tested across 12 product variants (e.g., floor, dish, toilet cleaners).
- Sold to JW Marriott, ISB, 10+ B2B clients; 22% net margins.

TRL: 6

- Prototype tested in a relevant environment with working production and third-party test validations.



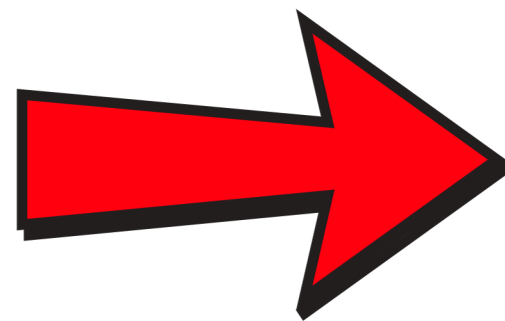
STATUS OF WORK ALREADY CARRIED OUT



Achieved stability at a PH level **exceeding 8** and temperatures **above 45 degree celcius.**

Formulation of **12 products** from same base.

Producing highly effective commercial grade cleaners from waste **CITRUS FRUIT PEELS** using our proprietary fermentation process.





No. of customers

B2B Segment

01



Hotels: 6

02



Academic
Institutions: 1

03



Cleaning
Companies: 1

D2C Segment

04



200 Customers
Indirectly(Surveys)
20+ Customers Directy

Technical Details of the Proposed Work to be Completed with SEED Support

FERMENTATION OPTIMIZATION & ENVIRONMENTAL MONITORING

FERMENTATION TIME REDUCTION

Objective: Apply Design of Experiments (DoE) to reduce bio enzyme fermentation time from 14–21 days to ≤ 7 days.

Method: Optimise microbial starter strains and fermentation parameters (pH, temperature, aeration, substrate load).

Goal: Increase production throughput without loss of enzymatic efficacy.

COD/BOD MONITORING

Objective: Conduct Chemical Oxygen Demand (COD) and Biochemical Oxygen Demand (BOD) tests on post-use grey water.

Method: Use APHA standard protocols to assess biodegradability and eco-safety of final products.

Goal: Empirically validate low environmental impact for regulatory and consumer confidence.

Technical Details of the Proposed Work to be Completed with SEED Support

MARKET-READY PRODUCT DEVELOPMENT

PRODUCT ENHANCEMENT & BENCHMARKING

- **Develop High-Performing Formulations:** Utilise bio enzymes as active agents.
- **Integrate Natural Ingredients:** Include stabilisers, fragrance agents, and surfactants for improved user appeal and efficacy.
- **Conduct comparative studies:** Compare performance with leading chemical products and eco-cleaners.

TARGET OUTCOMES

- **Achieve Category-Leading Performance:** Verify antimicrobial action, ensure pleasant fragrance, and attain eco-certification readiness.
- **Enable Large-Scale Production:** Scale-up to 1 ton/month through technical upgrades and streamlined processes.
- **Establish Cit-Peels as a Market Leader:** Focus on safe, sustainable, and scientifically validated cleaning solutions.



Our USP

1

**Toxic - free Cleaners,
natural and pleasant
fragrance, enhancing
user experience**

3

**Effective utilization of
waste : About 1/3rd of
every citrus fruit ends up
as waste**

2

**Clean homes leaves no
environmental stain**

4

**Kills 99.9% of Bacteria
tested by NABL
accredited labs**





Signs the market needs us !

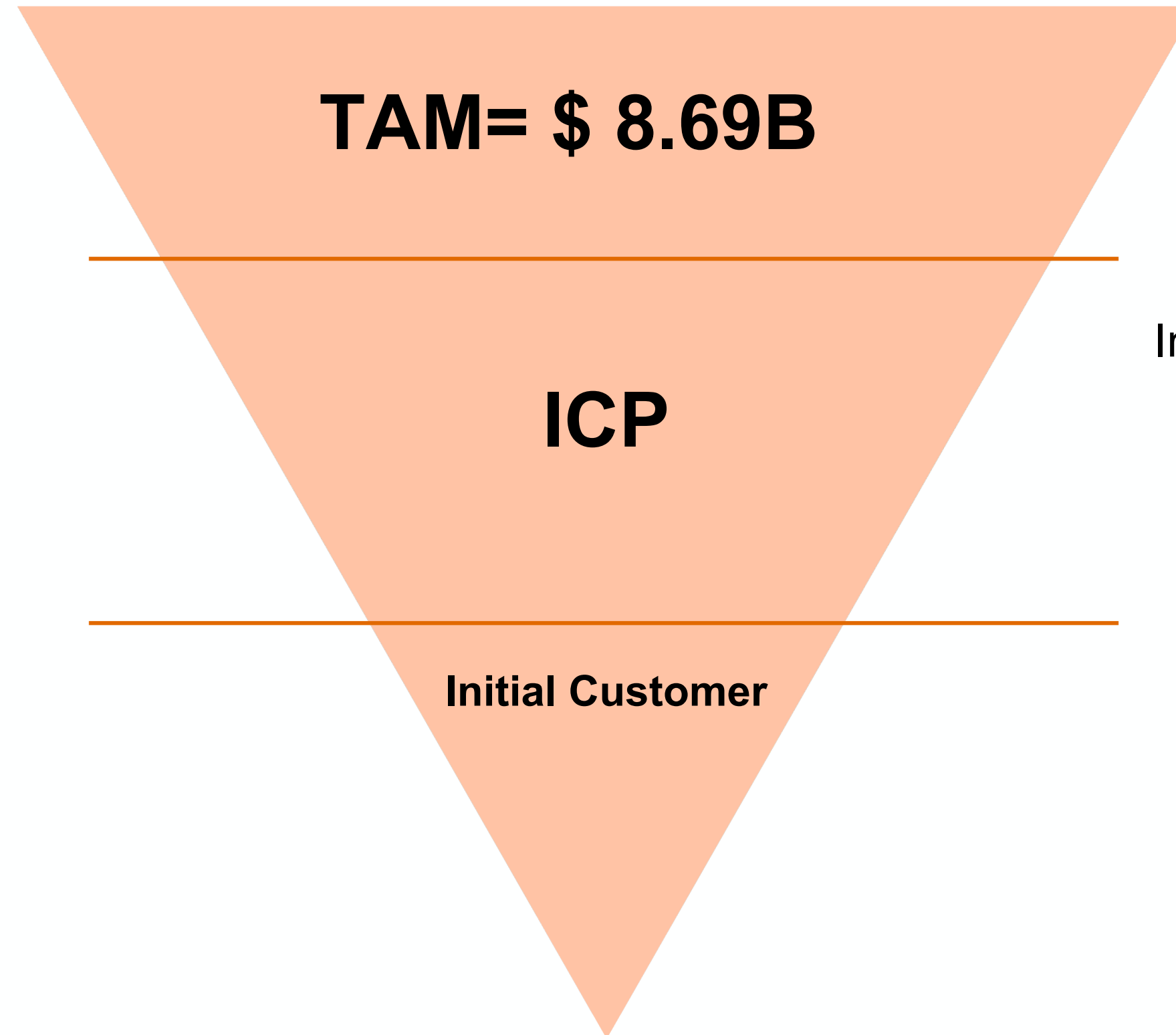
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Current Market Size : \$8.69
CAGR : 15.62%

2

Shift in consumer behaviour and

Market



India - 2.70 Million Households,
300 Million People,
Income : \$ 3k(per person)

India - 1 : The Consumer
Brand Testing and
Positioning Pool



Target Customer Segment

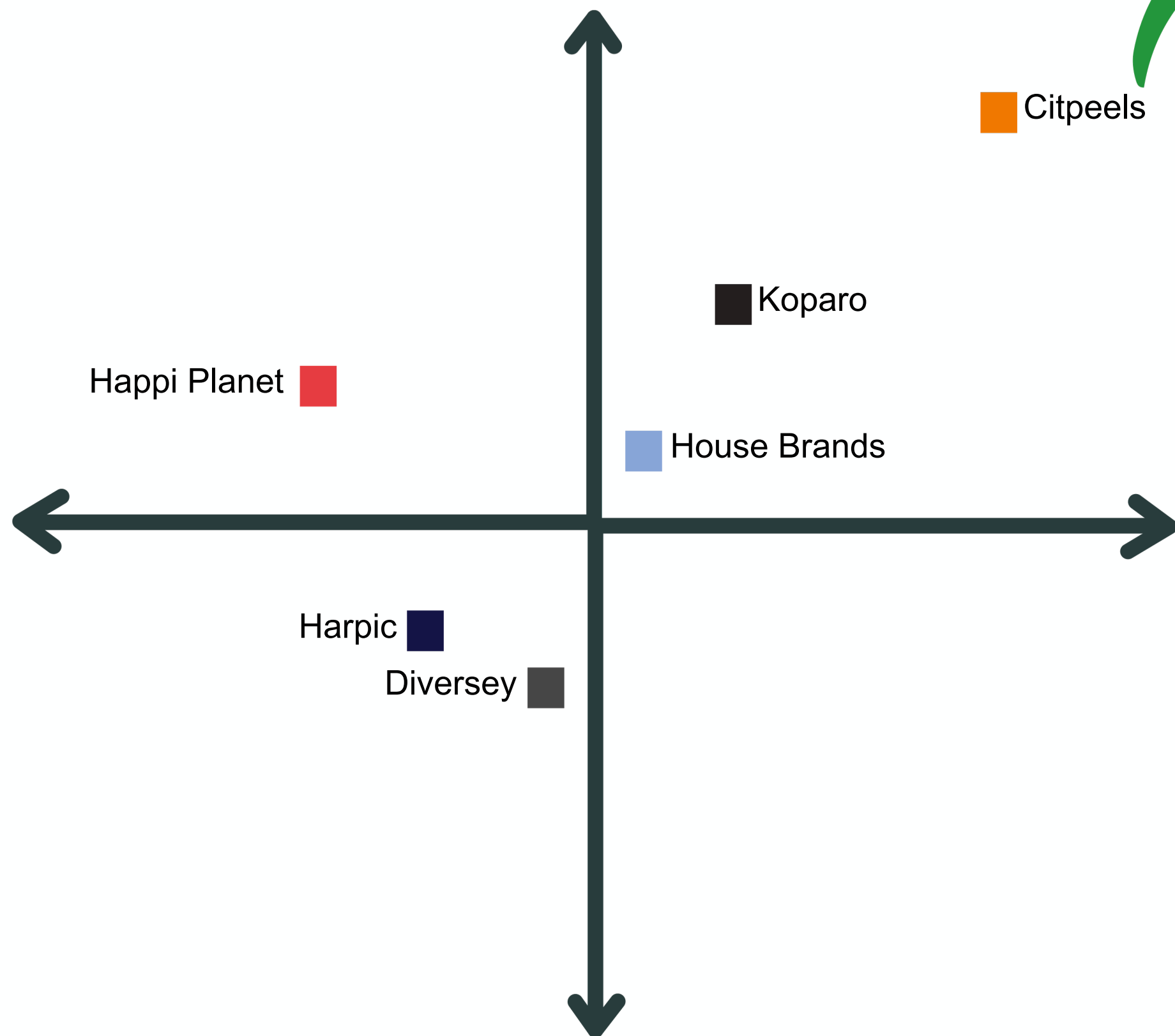
- **Government Hospitals, Hotel Chains:** Targeting this B2B segment which prefer cost-efficient solutions.
- **Educational Institutions:** Partnering with Schools, Colleges for Bulk orders.
- **Commercial Establishments:** Businesses, offices, and other commercial entities that require floor cleaning solutions for their workspaces.
- **Urban millennials and Gen Z:** Environmentally conscious individuals living in metropolitan cities.
- **Eco-conscious families:** Parents seeking safe and healthy cleaning solutions for their children and Pets.

•



Competition

Sustainability



WE STAND HERE

Affordability

Competitive Advantages, Market Differentiation, and Strategic Positioning

UNIQUE DIFFERENTIATORS

- **Up cycled Citrus Waste:**
Tackling 10 million tons of annual waste in India.
- **Proprietary Fermentation Process:**
Stable enzyme-rich cleaner with pH > 8
Heat-tolerant > 45°C.
- **Fully Biodegradable & Non-toxic:**
Safe for humans, pets, and the environment.
- **Residual Cleaning Power:**
Continues breaking down organic dirt post-application.
- **Proven B2B Traction:**

Interest from JW Marriott, ISB Mohali, cleaning contractors.

BACKED BY SCIENCE & IMPACT

- **Third-party testing:** antimicrobial and biodegradable test results validated.
- **Pilot Testing:** validated with real users and repeat B2B customers.
- **Early sales:** Partnerships

Novelty & Feasibility

Novelty

1. First-in-segment fermentation-based cleaner using citrus waste in India.
2. Developed through proprietary microbial culture and multi-stage fermentation delivering high enzymatic potency.
3. Achieves residual cleaning action — rare in current green cleaning products.
4. Converts agricultural waste into value-added, commercial-grade bioenzymes.

Feasibility

1. Technology Readiness Level (TRL-6): Prototype validated under real-world conditions.
2. Proven B2B demand: Sales to high-end clients (JW Marriott, ISB, institutions).
3. Fermentation process validated for pH > 8 and thermal stability > 45°C.
4. Pilot plant and small-batch production successfully executed.

IP Strategy & Regulatory Pathway



IP Strategy

1. Filing utility patent on proprietary fermentation protocol and enzyme formulation post laboratory validation.
2. Trademark application for brand identity is underway.

IP will cover:

1. Microbial blend and fermentation control
2. Product composition and applications
3. Unique use of citrus waste as primary substrate

Regulatory Approvals / Certifications

1. In process of acquiring:

Biodegradability Certification

Non-toxic Label Certification

Antimicrobial Efficacy Testing by NABL-accredited labs

2. Aligning formulation and claims with Ecomark and AYUSH standards for long-term product positioning.

Projected Time to Market & 3–5 Year Roadmap

Time to Market

- 1.MVP already validated through pilot batches and institutional B2B clients.
- 2.Projected market launch: Within 6–8 months of seed fund agreement.
- 3.Regulatory and certification completion: Parallel within Year 1.

3–5 Year Roadmap

YEAR	MILESTONES
Y1	Scale-upfermentation (200L),launch 3 SKUs, establish D2C & B2B distribution
Y2	Expand product line (12+ SKUs), achieve 1-ton/month capacity, nationwide distribution
Y3	Geographic expansion (Tier 1 & Tier 2 cities), enter international market
Y4-5	IP licensing, institutional contracts, partnerships with FMCGs or exports

Business Plan, Revenue Model & Target Customer

TARGET CUSTOMERS

- **B2B:** Hotels, hospitals, schools, cleaning contractors
- **D2C:** Eco-conscious households, Gen-Z/Millennials, pet parents
- **CSR Buyers:** Corporates with green mandates

REVENUE MODEL

- **B2B Sales (66.7%):** Focus on bulk orders and recurring institutional contracts
- **D2C (33.3%):** Website, retail, quick-commerce (Blinkit, Instamart), exhibitions
- **Margins:** Net margin ~22%; Product MRP=₹159, COGS ₹50

BUSINESS PLAN

- **Raw Material Sourcing:** Via juice vendors using waste citrus
- **Manufacturing Approach:** Lean manufacturing combined with proprietary fermentation
- **Marketing Strategy:** Awareness campaigns, trials, and B2B outreach



Unit Economics

Product	COGS (500ml)	MRP
Floor Cleaner	50	159
Utensils Cleaner	50	159
Laundry Wash	50	159
Room Freshners	124	249



BUSINESS MODEL

B2B

We are going to target different institutes, Govt. hospitals, Colleges and Schools, Hotels etc. We are having 22 percent net margins.

Cost Structure

- Raw material procurement (waste citrus peels from juice vendors).
- Manufacturing and packaging
- Marketing and distribution.

Key Partnerships

- Juice vendors for raw material sourcing.
- Logistics providers for distribution.
- Incubators and accelerators for mentorship and grants.

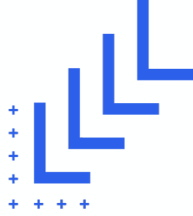
D2C

Direct selling from our *website*

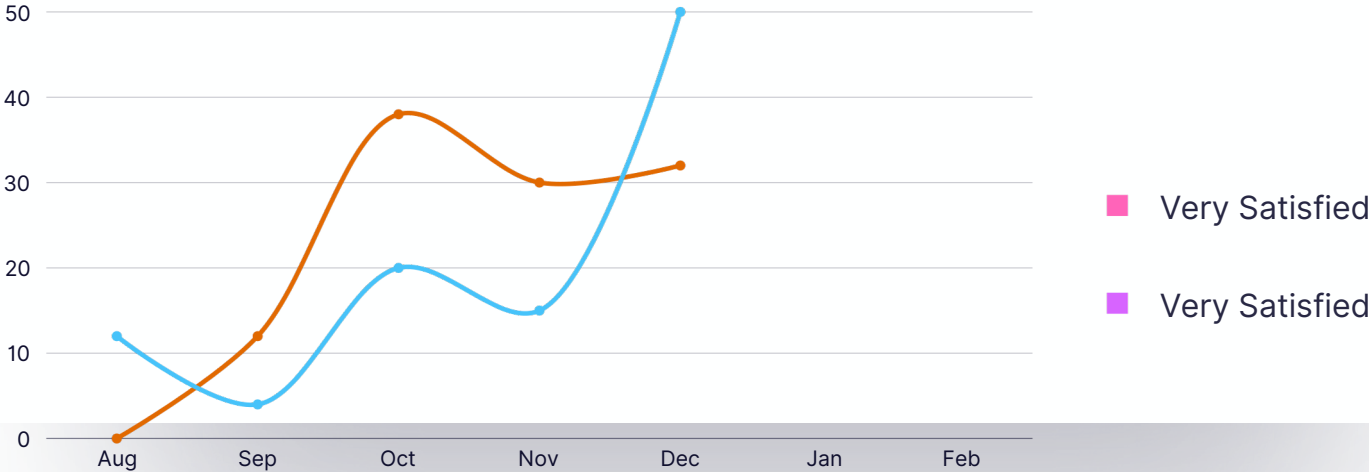
www.citpeels.in/home

Firstly we start from Chandigarh from retail stores, through quick commerce sites such as Blinkit, Swiggy Instamart. Here we have two Segments Premium And non-premium. We are also going to sell our products directly through different awareness marketing campaigns at various local exhibitions.

Sales(2024-25)

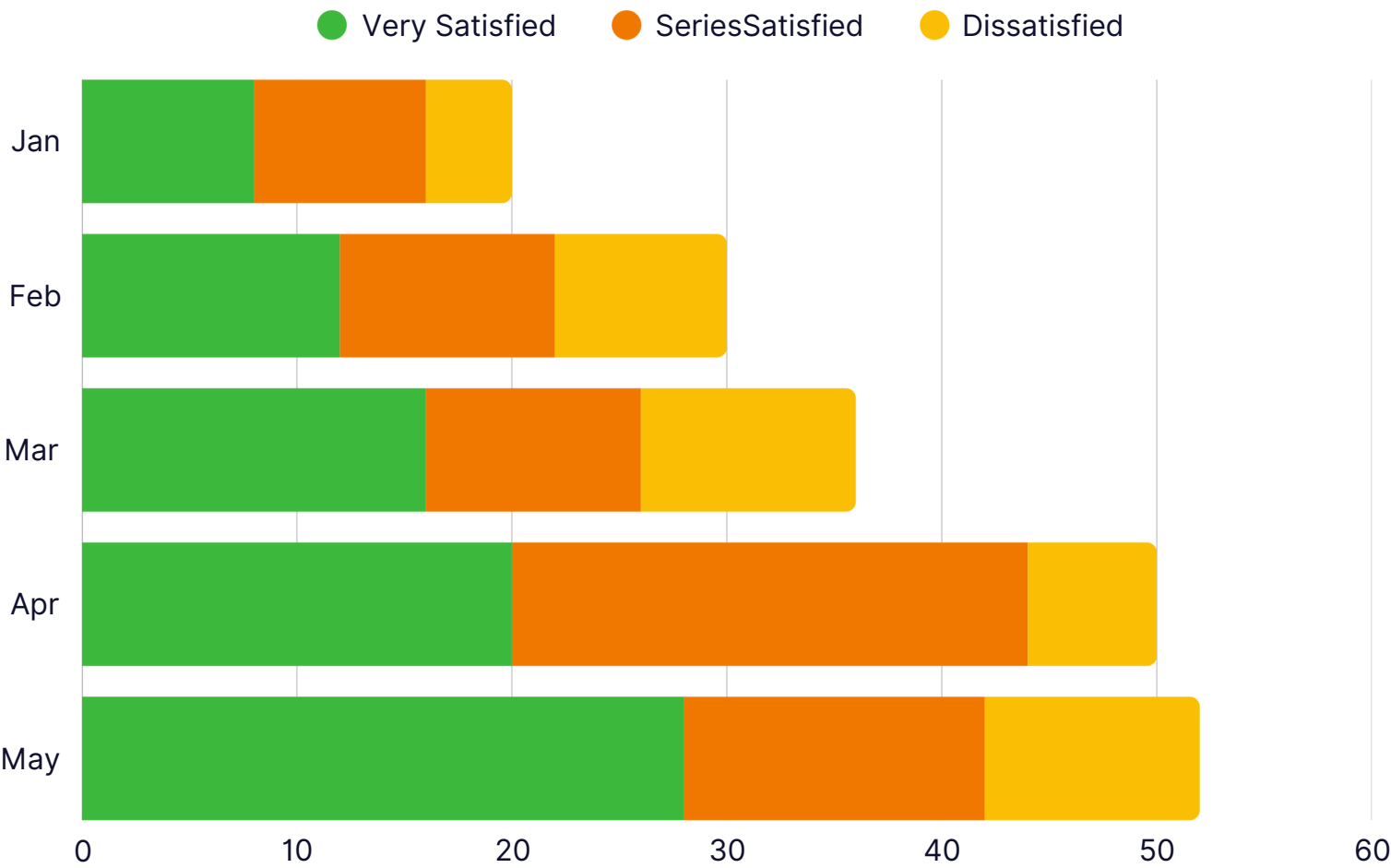


sales

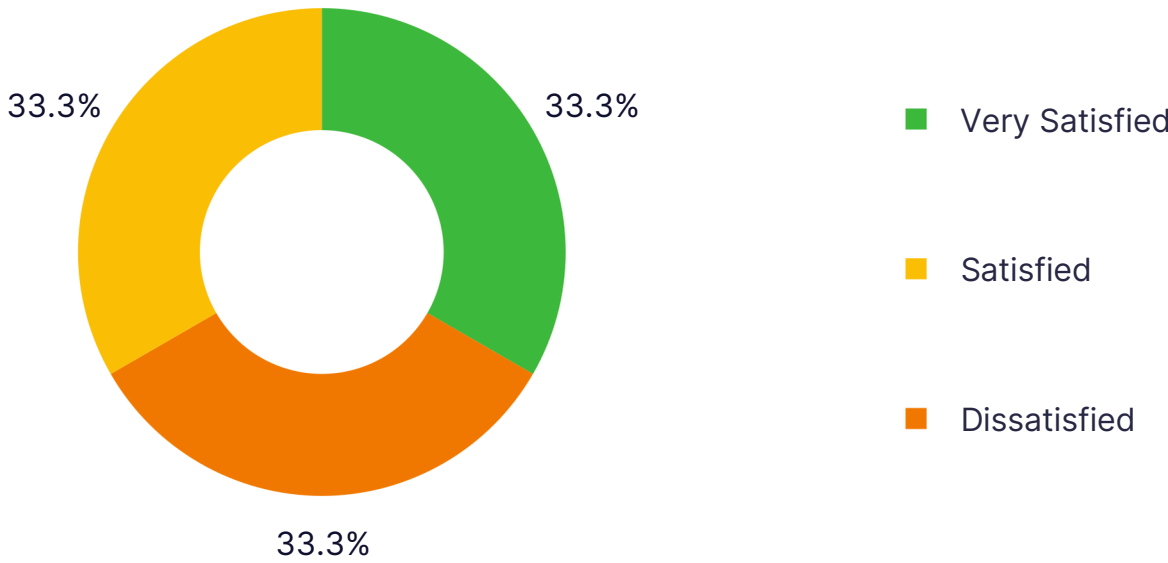


Data Visualizations

Employee Engagement Score

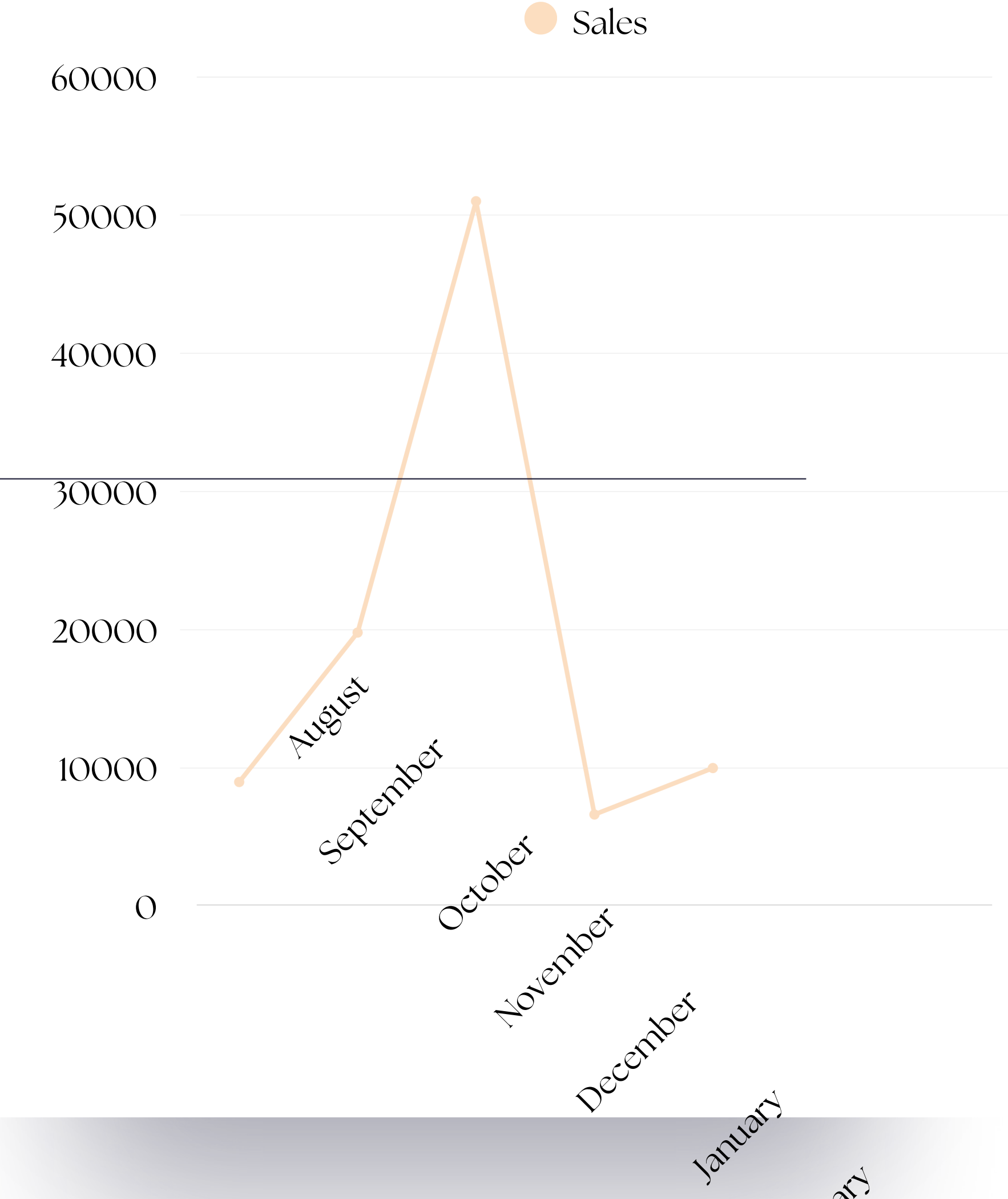


Customer Satisfaction

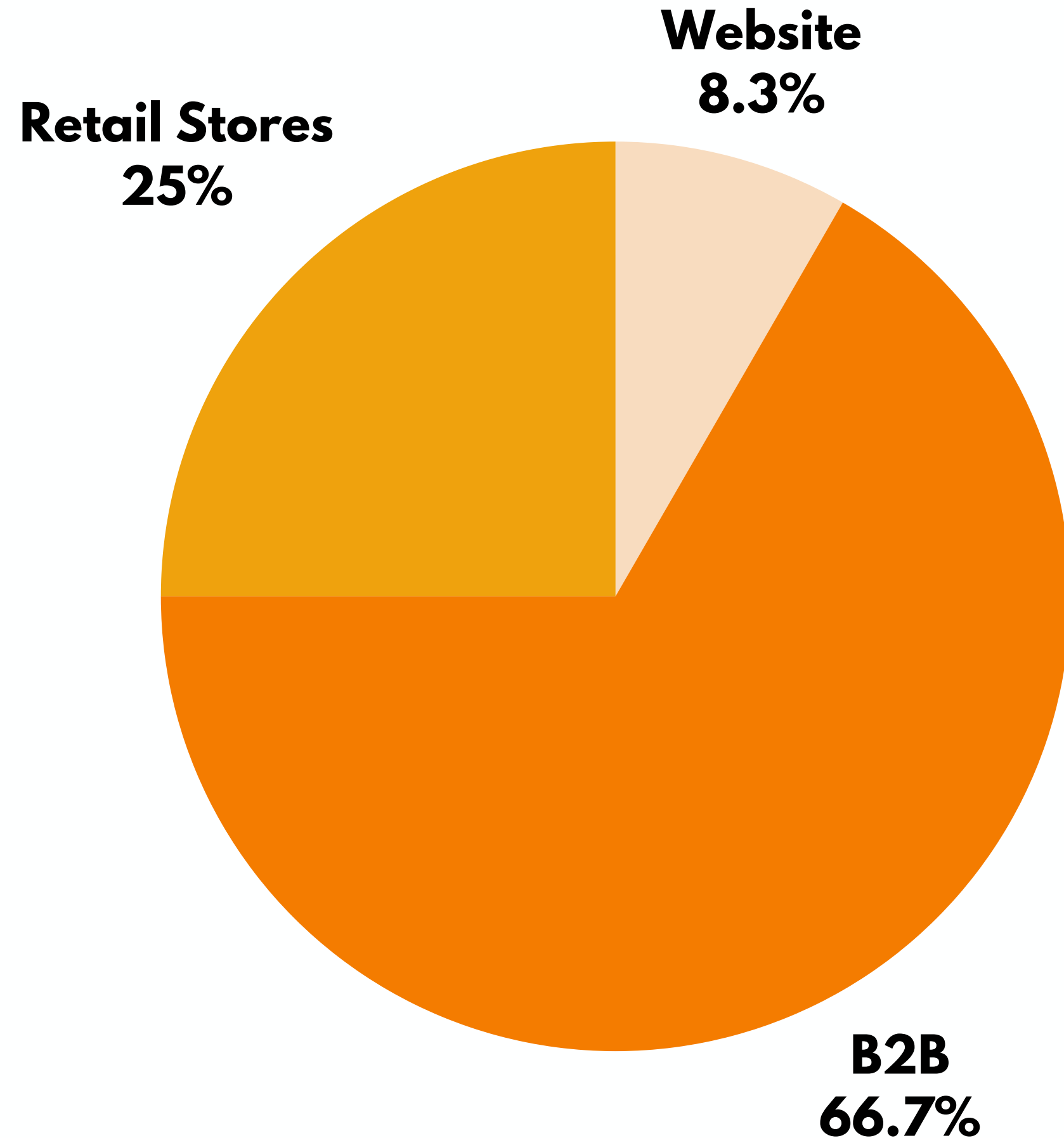


Sales(2024-25)

Customer Satisfaction



Sales Channel Share

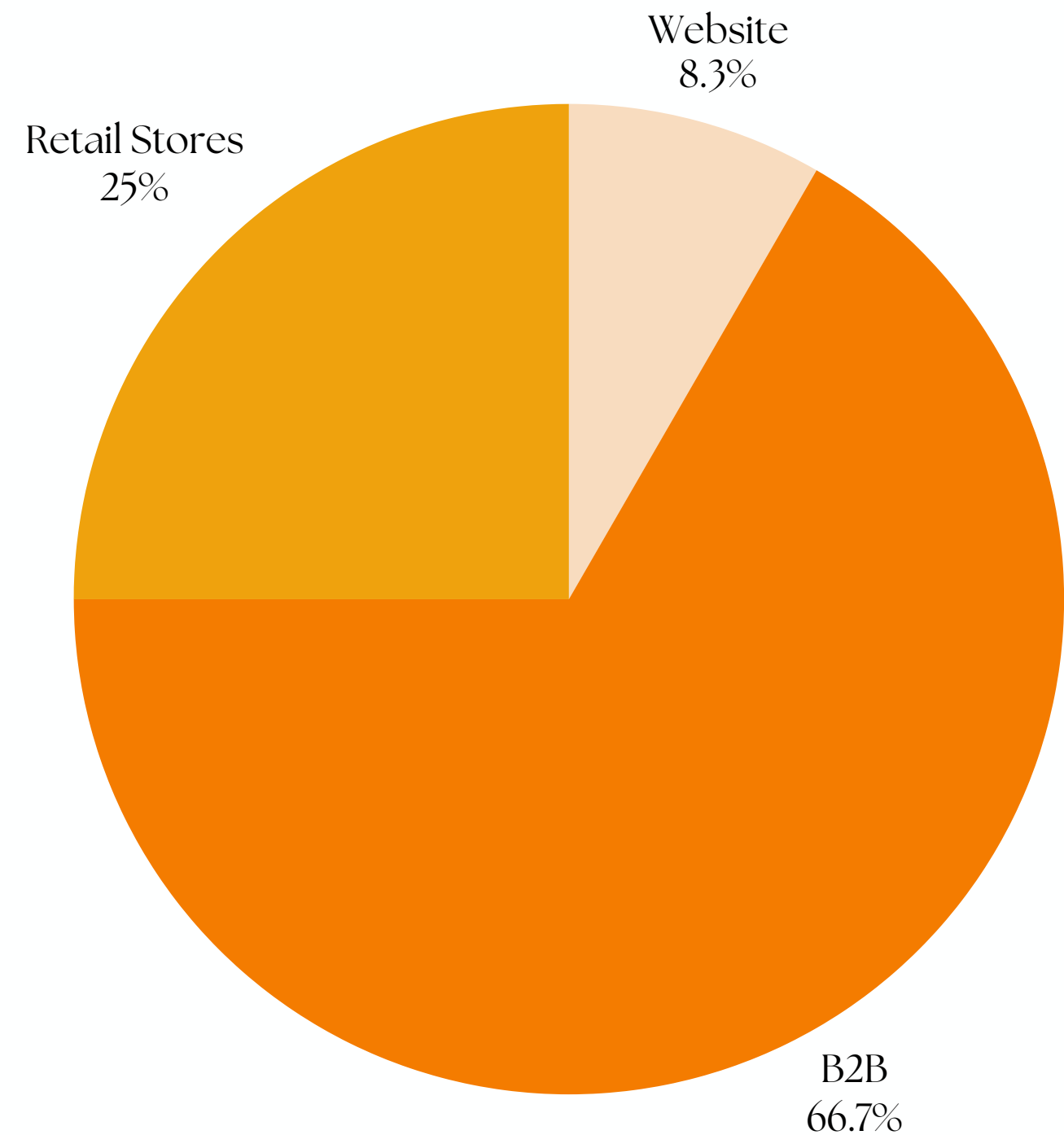


Major milestones

- Product placements at 3 shops
- Successfully done 4 white labelling deals out of which 2 gave repeat orders
- Partner with Cleaning Companies in Chd Tricity area
- Potential interest from 3 distributors
- Demo and interest from JW Marriot, ISB Mohali, IIT Ropar

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Sales Channel Share

Financial Risks & Market Validation

Potential Financial Risks

Risk: Delay in scaling production

Mitigation: Stepwise fermentation upgrade, maintain vendor flexibility

Risk: Raw material fluctuation

Mitigation: Long-term sourcing tie-ups with juice vendors

Risk: Slower market acceptance

Mitigation: Strong brand messaging, trials, and price benchmarking

Market Validation

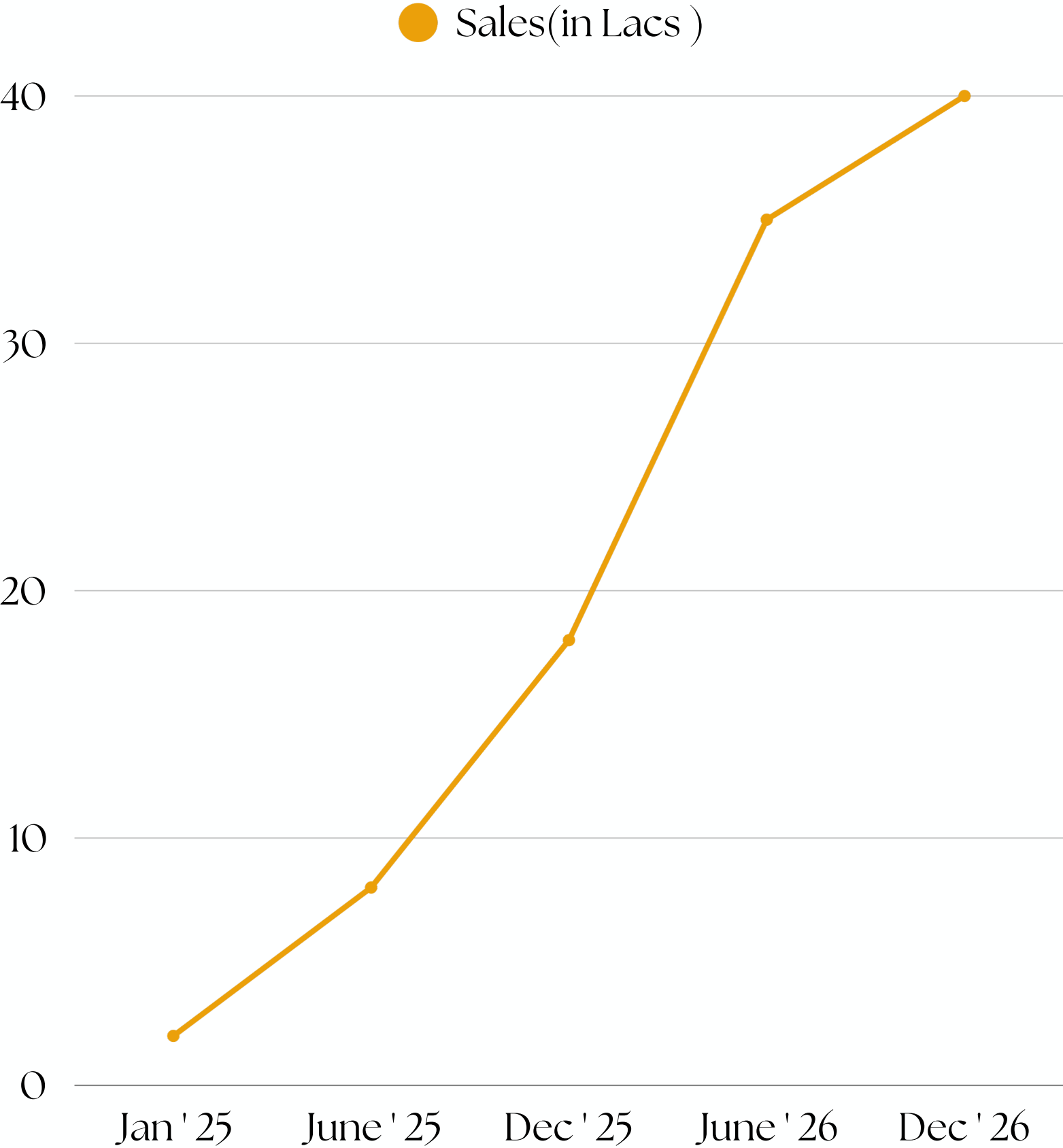
1. Pilot Sales: JW Marriott, ISB Mohali, 10+ B2B clients

2. Repeat Orders: From white-label clients and retail stores

3. Consumer Surveys: 200+ indirect users, 20+ direct D2C buyers

4. Certifications in progress: Biodegradability, toxicity, and efficacy

Future Sales Prospects



Cit-Peels Naturals – Year 1 Gantt Chart Work Plan

Activities	1	2	3	4	5	6	7	8	9	10	11	12
1. Lab-Scale Fermentation Optimization	✓	✓	✓									
- Strain selection, pH/temp trials	✓	✓										
- Nutrient/media optimization (DoE)		✓	✓	✓								
- Time reduction trials (≤7 days goal)			✓	✓	✓							
2. Pilot-Scale Trials (50L batch)				✓	✓	✓						
- Process scale-up from lab to pilot				✓	✓							
- Monitoring fermentation metrics				✓	✓	✓						
3. Product Development & Formulation					✓	✓	✓	✓				
- Variants (floor, dish, laundry, etc.)					✓	✓	✓					
- Additives, natural fragrance testing					✓	✓	✓	✓				

Cit-Peels Naturals – Year 1 Gantt Chart Work Plan

4. Certification & Environmental Testing							✓	✓	✓	✓	✓		
- Biodegradability & toxicity testing							✓	✓	✓	✓	✓		
- COD/BOD analysis on greywater							✓	✓	✓	✓	✓		
5. Product Stability & Shelf Life Testing								✓	✓	✓	✓	✓	✓
- Temp, pH, organoleptic stability								✓	✓	✓	✓	✓	✓
6. Scale-up Preparation (200L)									✓	✓	✓	✓	✓
- Equipment procurement & SOP drafting									✓	✓	✓	✓	✓
7. Branding & Packaging Development										✓	✓	✓	✓
- Label, packaging, safety & design compliance										✓	✓	✓	✓

Project Milestones & Timeline

Duration (Months)	Milestone Target
3 Months	☒ Completion of lab-scale fermentation optimization (strain selection, DoE trials, pH/temp profiling). ☒ Initiate small-batch COD/BOD testing on greywater.
6 Months	☒ Pilot-scale (50L) fermentation trials executed. ☒ Initial product formulations (3 SKUs: floor, dish, laundry) developed. ☒ Begin third-party certification process (biodegradability, toxicity). 
9 Months	☒ Completion of environmental impact testing (COD/BOD). ☒ Finalization of product performance parameters. ☒ Product stability testing (accelerated) initiated.
12 Months	☒ SOPs and QA/QC for scale-up finalized. ☒ Production-ready formulations for 5+ product variants. ☒ Begin branding and packaging design; prep for market launch.

15 Months
(optional extension)

✅ Commission 200L fermenter and begin semi-commercial batch production. ✅ Submit IP (patent filing) based on validated lab and pilot data.

18 Months
(optional extension)

✅ Launch pilot sales for scale production batch. ✅ Distribute to institutional partners and test market via D2C channels. ✅ Finalize logistics & onboarding of distribution partners.

Milestone Summary

M3: Lab-scale fermentation optimized

M6: Pilot batches completed; initial formulations ready

M9: Certifications and environmental test results available

M12: Scale-up SOPs, packaging, and product launch prep complete

AWARDS/ACCOMPLISHMENT

- ➡ **Won Sustainability competition by IIM Bangalore**
- ➡ **Best Idea award from BIRAC Panjab University**
- ➡ **Raised 10 lakh rupees from IIT Ropar and Angels**
- ➡ **BIRAC Eyuva fellowship of 2.5 lakh rupees**
- ➡ **Selected in ISB acceleration program on circular economy**





Future RnD Prospect

1

Carbon Calculation

2

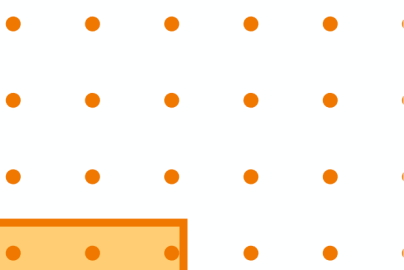
Impact on Water Bodies

3

Reducing fermentation time

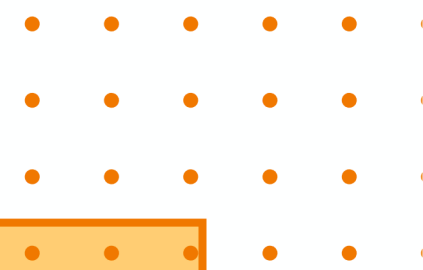


Proposed Budget Allocation(12 months)



Heads	Amount(in lakh)	Basis/Justification
Equipment	19.8	- Fermenters (50L & 200L): ₹18 Lakh - Stirrer Vessel: ₹0.8 Lakh - Filling & Capping Machine: ₹1 Lakh
Raw Materials	1.3	Procurement of Citrus waste,Jaggery,Packaging bottles , etc.
Testing & Certification	3	Biodegradability, toxicity, COD/BOD tests from NABL-certified labs
Consumables	0.7	Chemicals, lab media, pH strips, labware, etc. for fermentation and trials
Manpower	1.5	Technical intern(s), part-time lab assistant, field support (for 12 months)





Heads	Amount(in lakh)	Basis/Justification
Incubation Services	1.2	Access to lab facilities, shared equipment, and office space at TBIF
Travel	0.4	Local travel for vendor meetings, B2B demos, lab visits, and product testing
External Services	1.2	Design, packaging, branding agency fees, or part-time formulation consultants
Miscellaneous	2.4	Compliance, electricity, web hosting, domain, minor recurring expenses
Contingency	1.4	Reserve funds for unforeseen circumstances

Total Estimated Requirement: 28.9Lakhs



*Thank
You*

Let's #SkipTheChemicals together

